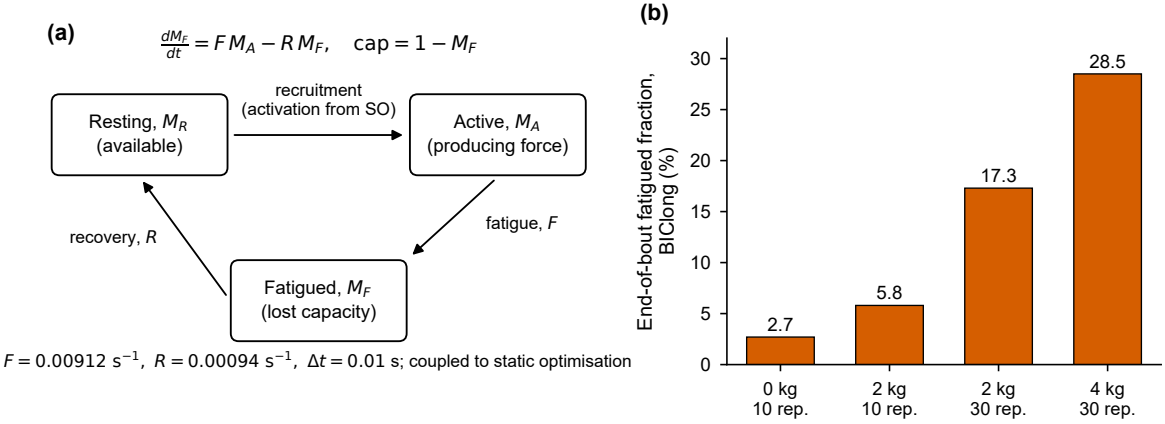
(a)
$\frac{dM_F}{dt} = F\,M_A - R\,M_F, \quad \text{cap} = 1 - M_F$
Resting, M_R (available)
recruitment (activation from SO)
Active, M_A (producing force)
fatigue, F
Fatigued, M_F (lost capacity)
recovery, R
$F = 0.00912\ \text{s}^{-1},\ R = 0.00094\ \text{s}^{-1},\ \Delta t = 0.01$ s; coupled to static optimisation
(b)
End-of-bout fatigued fraction, BICIong (%)
0
5
10
15
20
25
30
2.7
5.8
17.3
28.5
0 kg 10 rep.
2 kg 10 rep.
2 kg 30 rep.
4 kg 30 rep.